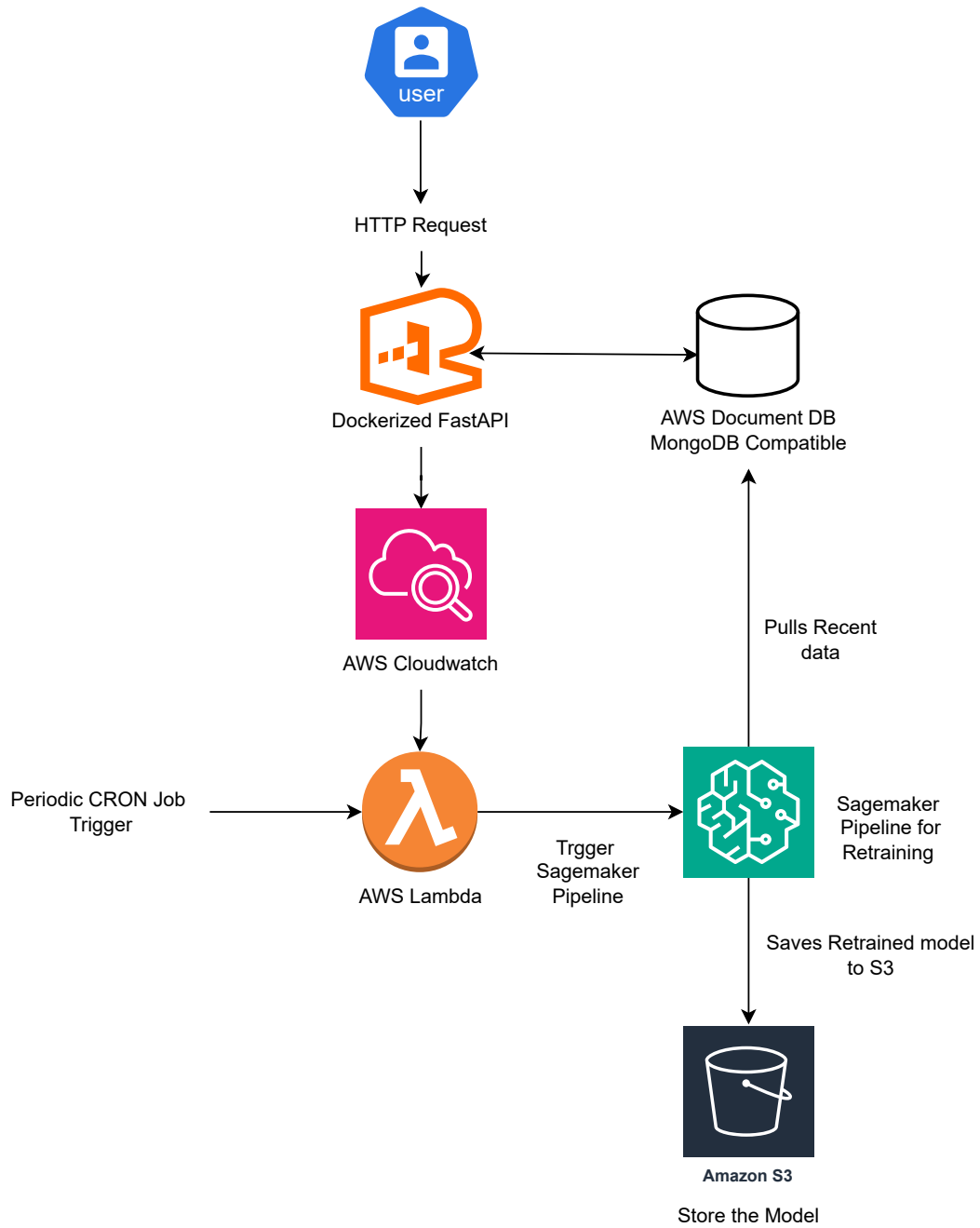




AWS Architecture Explanation (for MLE Project)

User / Client

- Sends **HTTP requests** to the deployed **FastAPI service**.
- These could be prediction requests or triggering data loading, etc.

Dockerized FastAPI

- The core **application backend**, containerized inside **Docker**.
- Hosted on an EC2 instance (or AWS ECS/Fargate optionally) running **FastAPI** server.
- Exposes API endpoints for:
 - Predicting results (Iris, News headlines)
 - Triggering retraining manually
 - Loading datasets, etc.

AWS DocumentDB

- A **MongoDB-compatible managed database** service from AWS.
- Stores:
 - Input samples (Iris or News data)
 - Any other necessary metadata or logs if needed.

Amazon S3 (Simple Storage Service)

- Used to **store the ML models**:
 - Initial trained model
 - Retrained versions after each cycle.
- Acts as a **central artifact store** for models.

AWS CloudWatch

- **Monitoring and logging** service.
- Monitors:
 - Application logs
 - Health of the FastAPI server
 - Cron job execution
 - Lambda execution success/failure
- Triggers alerts if services are down.

AWS Lambda

- A **serverless function** triggered periodically.
- Responsible for **automating the retraining** cycle.

- **Triggered by:** a **Periodic CRON Job** (using Amazon EventBridge scheduler).

Lambda's responsibilities:

1. Trigger a **SageMaker pipeline** for retraining.

◆ **Periodic CRON Trigger**

- Using **Amazon EventBridge Scheduler**.
- Schedules Lambda to run:
 - e.g., once per day/week.
- Automates retraining checks without manual intervention.

◆ **Amazon SageMaker**

- **Managed Machine Learning service** on AWS.
- **SageMaker Pipeline** automates:
 - Pulling latest data from MongoDB/S3
 - Re-training the model (Iris or News Classifier)
 - Evaluating the model (optional)
 - Saving the **retrained model** back into **Amazon S3**.

Note:

- **Load Balancer** (e.g., ALB) is NOT added because:
 - Single backend is enough for now.
 - No multi-node scaling requirement as per assignment.
- **Security** (IAM roles, security groups) would be needed in real production.